FINAL TECHNICAL REPORT

Freight Rate Prediction

Machine Learning Engineer Assessment

48,000	12,000	$131.88	31
Labeled development loads	Future validation loads	Best local holdout MAE	Fixed December dates

1. Executive Summary

Objective. Build a repeatable machine-learning workflow that predicts freight posted rates for 12,000 future loads, while validating the approach on a realistic future holdout and producing the fixed December chart required by the assessment scorer.

Approach. The workflow performs data-quality repair without deleting rows, uses time-aware validation, engineers temporal/geographic/market interaction features, benchmarks several models, retrains the selected candidate on all labeled development data, and validates the final output schema before export.

Main result. HistGradientBoosting was the strongest model family. The absolute-error configuration achieved the lowest local holdout MAE ($131.88), which is consistent with its robustness to a strongly right-skewed target and extreme rates. However, in the fixed December sensitivity test it produced a much flatter day-to-day profile. The squared-error HGB variant had slightly weaker holdout metrics (MAE $163.54) but showed clearer response to temporal and daily market-context features, so it was selected as the final candidate for the submission.

Important model-selection note

Absolute-error HGB was the metric winner on the October holdout, but model selection was not based on MAE alone. In the fixed December scenario, the absolute-loss model showed limited day-to-day responsiveness, while squared-error HGB produced clearer temporal variation with shipment attributes held constant. The squared-error version was therefore used for the final candidate. This does not imply that the absolute-loss model literally ignored date features; rather, its response was too flat for the fixed-scenario diagnostic.

2. Data Understanding and Exploratory Findings

- Development data contains 48,000 labeled loads from January through October 2025; the future validation set contains 12,000 loads from November through December 2025.
- The prediction target is posted_rate. The target is strongly right-skewed: median about $2,031, P99 about $5,973, and maximum $25,533.
- Distance is the strongest simple numeric relationship with posted_rate, with Pearson correlation of approximately 0.909.
- Equipment types have different average rate levels: Reefer is highest on average, followed by Flatbed and Dry Van.
- The future validation distribution is not identical to development data. Mean market_index shifts from about 1.083 in development to about 0.927 in validation.

- Validation contains 8 cities not seen in development; 1,447 of 12,000 loads (12.1%) touch at least one unseen city. There are also 736 unseen exact routes covering 1,461 validation loads.

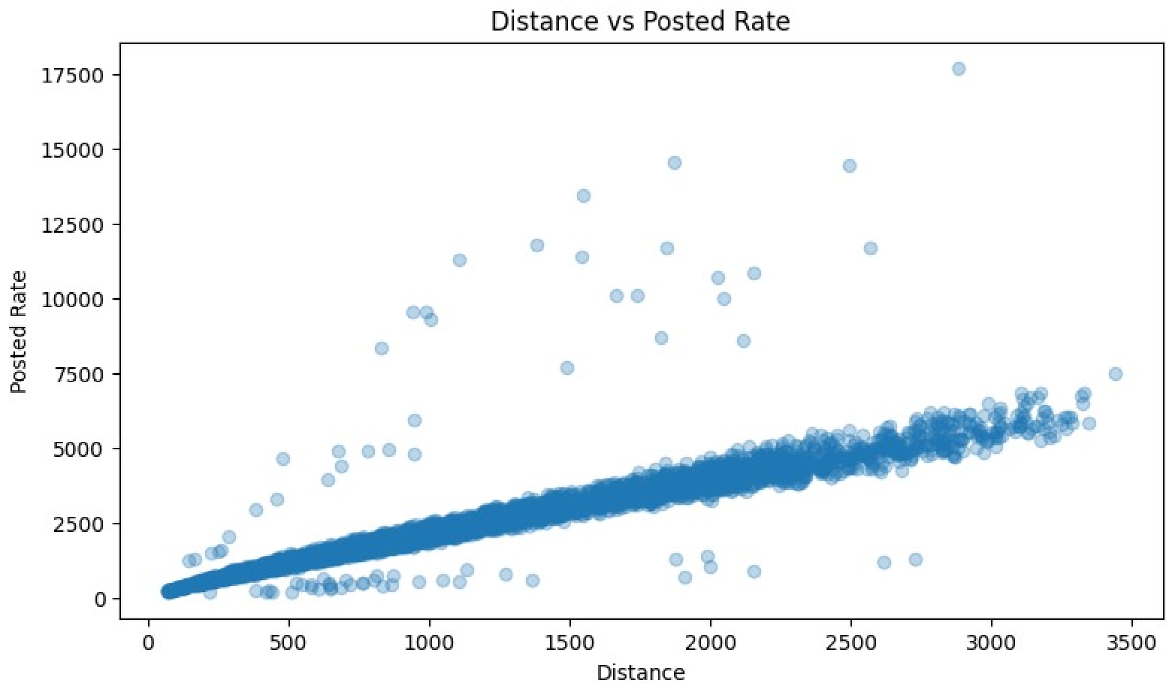

Figure 1. Distance vs. posted rate from the notebook EDA. The strong near-linear structure supports distance as the dominant simple predictor, while the scattered extreme observations illustrate the heavy tail.

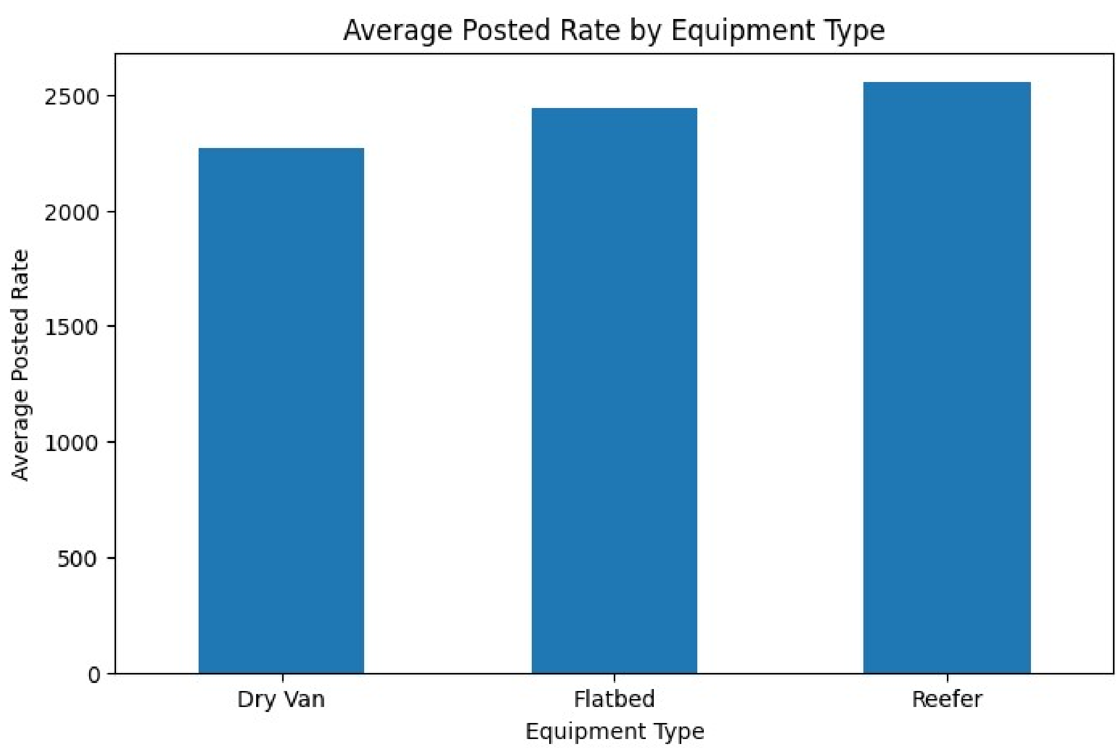

Figure 2. Average posted rate by equipment type. Reefer has the highest average rate, followed by Flatbed and Dry Van, supporting equipment as a useful categorical feature.

3. Data Quality and Cleaning

Issue	Development	Final validation	Treatment
Missing weight	300	165	Impute by equipment-specific median
Negative weight	292	145	Convert to absolute value; preserve audit flag
Missing market_index	374	249	Impute with development median 1.0558
Duplicate load IDs / rows	None found	None reported	No removal required

Outlier policy. No target rows were capped or removed. The extreme posted rates were retained because there was no evidence that they were data-entry errors, and deleting them would remove real tail-market behavior. Negative weight values were treated differently because their magnitudes looked like ordinary weights with a corrupted sign.

Auditability. Three binary quality flags were retained: weight_missing_flag, weight_negative_flag, and market_index_missing_flag. This allows the model to distinguish repaired observations from ordinary ones instead of hiding the repair step.

4. Validation Strategy

Why time-based validation? The assessment predicts future loads, so a random split would mix earlier and later market states and could produce an overly optimistic score. The local holdout therefore uses the most recent labeled month as a future-like test window.

Split	Date range	Rows
Training	2025-01-01 to 2025-09-30	43,147
Local validation	2025-10-01 to 2025-10-31	4,853
Final assessment set	2025-11-01 to 2025-12-31	12,000

Leakage control

Cleaning statistics and preprocessing are learned from development/training data and then applied forward. Unknown categories are handled explicitly rather than forcing validation information back into training.

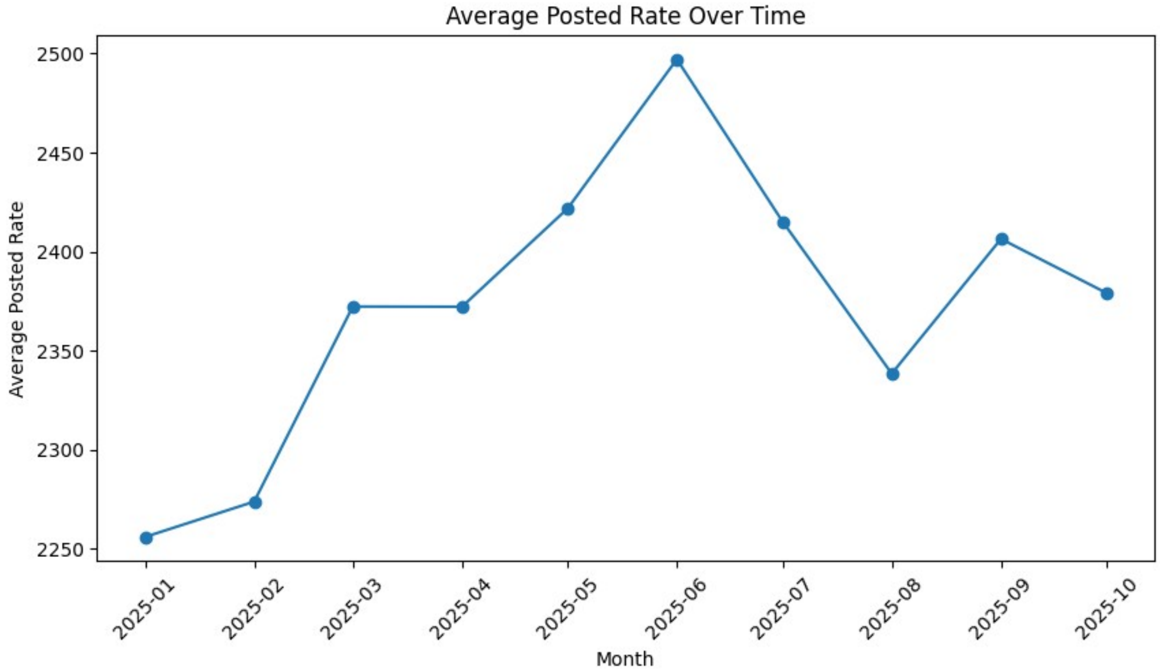

Figure 3. Average posted rate over time from the notebook EDA. The monthly level changes across 2025, supporting a future-like time-based validation split instead of a random split.

5. Feature Engineering and Preprocessing

The same feature logic is applied consistently to training, local validation, final validation, and the December scorer scenario. Key derived variables include:

- Time: month, day_of_week, and days_since_start.
- Geography: absolute latitude and longitude differences between pickup and delivery.
- Market interactions: distance_market = distance x market_index and distance_quote = distance x quote_signal.
- Categorical context: pickup city, delivery city, and equipment type.
- Original numeric inputs: distance, weight, coordinates, market_index, quote_signal, and data-quality flags.

Encoding. For HGB, categorical variables are ordinal-encoded with an explicit unknown value (-1), while numeric features pass through without scaling. load_id is excluded from modeling so the model cannot memorize record identifiers.

6. Model Benchmark and Error Analysis

Rank	Model	MAE	RMSE	MAE improvement
1	HGB - Absolute Error Loss	$131.88	$648.87	88.82%
2	HGB - Squared Error Loss	$163.54	$663.59	86.14%
3	Ridge Regression	$171.01	$674.67	85.51%
4	Random Forest	$410.30	$800.58	65.22%
5	Dummy Baseline (Median)	$1,179.80	$1,528.57	Benchmark

Interpretation. The nonlinear HGB models clearly outperform both the median baseline and Random Forest. Absolute-error loss achieves the best local MAE because it is less dominated by the extreme right tail. Squared-error HGB remains close on the holdout metrics, but places more weight on larger residuals and, in the fixed-scenario diagnostic, produced stronger day-to-day variation.

Why MAE and RMSE tell different stories

The top 1% of validation errors contributed approximately 89.9% of total squared error. A small number of very high freight rates were heavily underpredicted, so RMSE is much larger than MAE. This is why both metrics were monitored instead of relying on a single score.

6.1 Model Selection Trade-off: Metrics vs. Temporal Responsiveness

Metric winner. HGB with absolute-error loss achieved the best October holdout MAE ($131.88) and RMSE ($648.87). This result is consistent with the dataset: posted_rate is strongly right-skewed, and a small number of extreme observations dominate squared error. Absolute loss reduces the influence of those tail observations and therefore performs very well on aggregate validation metrics.

Fixed-scenario diagnostic. The model was also inspected on the required December fixed-load scenario, where route, distance, equipment, and weight remain constant and only the date/context changes. In candidate testing, the absolute-loss configuration produced a comparatively flat profile across dates, indicating limited practical sensitivity to the temporal signal in this diagnostic.

Final decision. The squared-error HGB had slightly weaker local metrics (MAE $163.54; RMSE $663.59), but its December predictions showed clearer day-to-day movement with the physical shipment definition fixed. This suggested stronger use of the engineered date/day and market-context features. For that reason, squared-error HGB was selected as the final submission candidate. The choice is therefore a deliberate trade-off between robust aggregate error metrics and temporal responsiveness, not a claim that the absolute-loss model completely ignored date features.

Configuration	Local validation	December fixed-scenario behavior
HGB - Absolute Error	Best MAE: $131.88	More robust to outliers, but comparatively flat across dates
HGB - Squared Error	MAE: $163.54	Clearer day-to-day variation; selected final candidate

7. Final Candidate Training and Output Validation

Final training. After local model comparison and the fixed-scenario responsiveness check, the squared-error HistGradientBoosting candidate was refit on all 48,000 labeled development rows before generating predictions for every load in the 12,000-row assessment validation set.

Output checks. The saved validation_predictions.csv contains exactly load_id and predicted_rate. The notebook checks that all 12,000 predictions are present, load IDs remain aligned with the provided validation file, and there are no missing, duplicate-ID, or non-positive predictions.

Minimum prediction	Mean prediction	Maximum prediction
$329.21	**$2,426.14**	**$7,930.07**

8. Fixed December Prediction Chart

Scenario. The scorer uses a fixed Lexington to Fort Wayne load: 360 miles, Dry Van, 32,000 lb, across the 31 dates in December 2025. The resulting series is deliberately evaluated as a fixed operational scenario so that date/context sensitivity can be inspected without route or equipment changes.

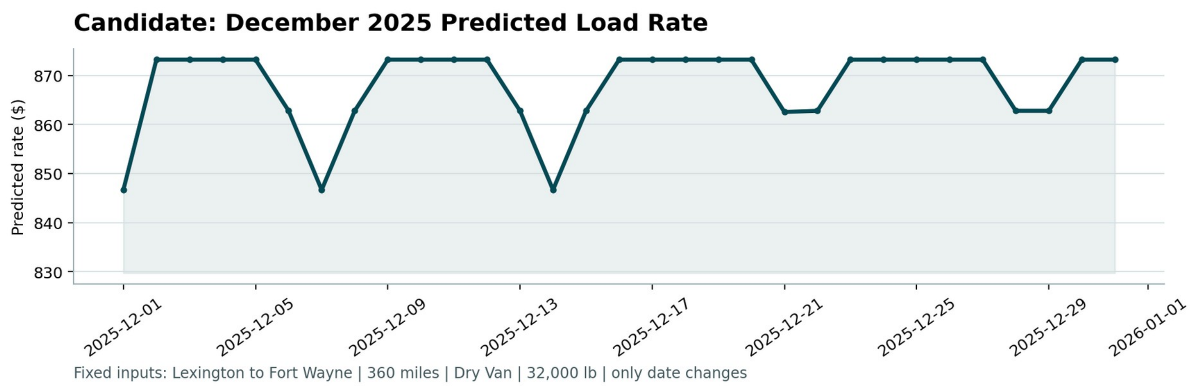

Figure 4. Candidate December 2025 predicted load rate - fixed Lexington to Fort Wayne scenario.

Rows	Minimum	Mean	Maximum
31	**$846.68**	**$867.94**	**$873.21**

Chart interpretation. The final uploaded scorer series ranges from approximately $846.68 to $873.21, with a mean of $867.94. It is not a flat line: the fixed load receives lower predictions on selected dates and an upper plateau on many dates. Because route, distance, equipment, and weight are held constant, the visible movement is evidence that the selected squared-error pipeline responds to the changing date/daily market context. This fixed-scenario behavior was one reason it was preferred over the lower-MAE absolute-loss configuration.

9. Reproducibility and Scorer Validation

The repository is designed so the full workflow can be reproduced from the notebook and the supplied scorer. The final scorer command is:

```
python score.py \
  --predictions validation_predictions.csv \
  --december-predictions data/december_chart_inputs.csv \
  --output-dir scorer_results
```

The final scorer should confirm:

- 12,000 final validation predictions are valid.
- 31 fixed December predictions are valid.
- The December chart is created successfully in scorer_results/.
- Final hidden assessment metrics are calculated by Spotter after submission.

10. Conclusion

The project demonstrates an end-to-end freight-rate ML workflow with realistic future validation, explicit data-quality handling, reproducible feature engineering, model benchmarking, output validation, and scorer-compatible artifacts. HGB with absolute-error loss was the strongest model by local MAE, but the final squared-error HGB candidate was chosen after an additional fixed-scenario diagnostic showed better temporal responsiveness. The final decision therefore balances robust holdout metrics against the practical requirement that a fixed shipment respond meaningfully to date and market context. The December scorer chart confirms that the final pipeline produces date-sensitive predictions rather than a constant output.

Final assessment artifacts

validation_predictions.csv (12,000 rows) | data/december_chart_inputs.csv (31 rows) | scorer_results/candidate_december.png | notebook + README with run instructions

Methodology note: Figures 1-3 are direct notebook EDA outputs and Figure 4 is the required final December scorer chart. The absolute-vs-squared fixed-scenario comparison is an empirical candidate-selection observation from the project workflow; local validation metrics are not the hidden final assessment score.